

University of Chittagong
Department of Computer Science and Engineering
7th Semester B.Sc. (Engg.) Final Examination-2024
Course Code: CSE-717
Course Title: Information Security

Total Marks: 54 Marks

Time: 4.00 hours

[Answer any *three* questions from each of the *Section-A* and *Section-B*; a Separate answer script must be used for Section-A and Section-B. Figures in the right-hand margin indicate full marks]

Section-A

- 1.a) Briefly describe symmetric and asymmetric cryptography with an example. Point out the reason why asymmetric cryptography is useful for e-commerce. 3
 - b) What is handshaking? Briefly describe how message-digest works. 2
 - c) An information system can be in one of two states: Secure (S) or Compromised (C). If the system is Secure during a security assessment cycle (active patching, effective controls), it remains Secure with probability 0.4. If the system is Compromised during an assessment cycle (failed controls, unpatched vulnerabilities), incident-response actions return it to Secure with probability 0.3. Determine the steady-state probabilities of Secure (S) and Compromised (C) states, and evaluate whether the system is more likely to be secure or compromised in the long run. 3
 - d) How the concept of finite field are used in cryptography? 1
 - 2.a) Encrypt the message "CRYPTO" first with a Caesar cipher shift of 5. 1.5
 - b) A message was encrypted using a Caesar cipher with the encryption function $f(p) = (p + 9) \bmod 26$. The ciphertext received is "ONXNG". Determine the original plaintext. 1.5
 - c) Encrypt the message "DATA" using the RSA cryptosystem with $p = 61$, $q = 53$, and $e = 17$. 3
 - d) An RSA-encrypted message consists of the ciphertext block $c=17$. The RSA system parameters are: $p = 61$, $q = 53$, and $e = 17$. Decrypt the ciphertext and find the plaintext letter. 3
 - 3.a) Explain how ShiftRows and MixColumns together provide diffusion in AES. Given the state matrix (bytes in hexadecimal) shown on the left, perform the ShiftRows transformation. 4
- State =

$$\begin{bmatrix} d4 & e0 & b8 & 1e \\ 27 & bf & b4 & 41 \\ 11 & 98 & 5d & 52 \\ ae & f1 & e5 & 30 \end{bmatrix}$$
- b) A company maintains a block chain based distributed ledger for its digital transactions. Each new block contains the hash of the previous block, forming a chain. To add a block, participants must solve a Proof of Work puzzle, making tampering computationally difficult. Now, explain how blockchain ensures security, specifically in terms of hashing and Proof of Work. 3
 - c) Draw the advanced Encryption Standard (AES) encryption block diagram. 2

None for both

- 4.a) Alice is using a toy version of the DSS signature scheme with a prime modulus $P=47$ and generator $g=2$ of order $q=23$. By accident, Bob generates signature for two different messages with the same per-message random number K . The hash codes of two signed messages are 2 and 3 and signatures are $(4, 21)$ and $(4, 19)$. Compute the private key used by Bob. 4
- b) Describe how digital certificates, certificate authorities, and SSL work together to exchange data on the web securely and reliably with an illustrative example? 3
- c) Using the extended Euclidian algorithm, find the multiplicative inverse of $550 \bmod 1769$. 2

Section-B

- 5.a) Describe the security issues you need to consider while launching an online shopping portal. 3
- b) What is the secure electronic transaction (SET)? Discuss the SET process used in an e-commerce transaction. 3
- c) Explain the architecture of the IPsec and discuss the role of security associations. 3
- 6.a) Given an elliptic curve $E : y^2 = x^3 + 2x + 3$ over $\mathbb{F}_{97}$, compute $2P$ for the point $P = (3, 6)$. 3
- b) Explain why ECC and RSA are more vulnerable than symmetric algorithms to quantum attacks. 2
- c) Write short note on Denial of Service (DoS) attack and SQL Injection attack. 2
- d) What do understand by avalanche effect? Write down two families of attacks in DES. 2
- 7.a) Eve and Mallory are choosing which cybersecurity training to attend: Penetration Testing (PT) or Network Defense (ND). Their satisfaction (payoff) depends on both their choices: they prefer attending the same training together, but each has slightly different preferences. The payoff table shows (Eve's payoff, Mallory's payoff): 3

		Mallory	
Eve		PT	ND
	PT	(4, 3)	(1, 2)
	ND	(2, 1)	(3, 4)

- i) Identify all Nash equilibria in this scenario.
- ii) Does Eve have a strictly dominant strategy? Does Mallory?
- b) Solve the congruence equation $195X \equiv 23 \pmod{968}$ 3
- c) A hash table of size $m = 13$ uses double hashing. Primary hash function: 3
- $h_1(k) = k \bmod 13$ and secondary hash function: $h_2(k) = 7 + (k \bmod 7)$.
- Now, insert the following keys in order: $\{8, 47, 22, 44, 59, 32\}$

- 8.a) What is the firewall? Write down the advantages and disadvantages of packet filters. 1+2
- b) What is packet sniffing? How does it differ from a firewall? 2
- c) What are the port scanning techniques? Write down the usage of packet sniffers. 1+2
- d) Draw the generic model of Digital Signature process. 1

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Department of Computer Science and Engineering
7th Semester B.Sc. (Engg.) Final Examination-2023
Course Code: CSE-7197 Course Title: Cryptography and Network Security
 Total Marks: 54 Marks Time: 4.00 hours

[Answer any **three** questions from each of the **Section-A** and **Section-B**; a Separate answer script must be used for Section-A and Section-B. Figures in the right-hand margin indicate full marks]

Section-A

- 1.a) Using the following Playfair matrix, encrypt the following message:

See you at CSECU

M	F	H	I/J	K
U	N	O	P	Q
Z	V	W	X	Y
E	L	A	R	G
D	S	T	B	C

- b) Using Fermat's theorem, find $3^{202} \bmod 11$. 2.5
 c) Use, Euler's theorem to find a number m in between 0 and 9 such that m is congruent to 7^{1000} modulo 10. 2.5

- 2.a) What is a digital signature? Briefly describe the digital signature process of public-key cryptography to ensure security over the internet. 3

- b) Describe how digital certificates, certificate authorities, and SSL work together to exchange data on the web securely and reliably with an illustrative example? 4

- c) Explain briefly how the following attacks occur: 2
 i) Phishing attack.
 ii) Denial of service attack.
 iii) SQL injection attack.
 iv) Clop ransomware.

3. a) Draw a block diagram of 3-DES. Write down a simple equation to represent DES encryption for 3-DES. 3

- b) What do you understand by avalanche effect? Write down the two families of attacks in DES. 2

- c) Draw the advanced encryption standard (AES) encryption block diagram. 4

4. a) State the Chinese remainder theorem. Solve the following systems of congruence by using the Chinese remainder theorem. 5

$$X \equiv 2 \pmod{3}$$

$$X \equiv 4 \pmod{5}$$

$$X \equiv 6 \pmod{7}$$

- b) Using game theory, design a strategy to identify attackers in a mobile social network. 3

- c) Discuss about two atomic operations proposed by Shannon in security. 1

Section-B

- 5.a) Draw the details of F-function in DES. 2.5
 b) Enumerate the properties of a good hash function. 2.5
 c) Briefly describe the types of attacks that make the digital signature method vulnerable. 3
 d) In the RSA public-key encryption scheme, each user has a public key, e , and a private key, d . Suppose Bob leaks his private key. Rather than generating a new modulus, he decides to generate a new public and a new private key. Is this safe? 1

6. a) Describe the security issues you need to consider while launching an online shopping portal. 2
 b) What is handshaking? Briefly describe how message-digest works. 3
 c) Consider a network system that can either be in a secure state or an insecure state. The network system periodically undergoes patches to maintain security. However, there's a risk that the system may be vulnerable to attacks at any given time. 4
 i) When the system remains secure because the patching mechanism is effective, and there are no new vulnerabilities, there is an 80% chance that the system remains secure after a security check.
 ii) When no patch is applied or the patching mechanism fails, the system remains insecure and there is a 60% chance that the system remains secure after a security check.

Determine the long-term behaviour of this network system, whether it is secure or insecure.

7. a) What is the secret message produced from the message "CSECU" using the Caesar cipher with the encryption function $f(p) = (p + 12) \bmod 26$? 2
 b) Decrypt the ciphertext message "CHATTOGRAM" that was encrypted using the Caesar cipher encryption technique with shift $k = 7$. 2
 c) Encrypt the message "CU" using the RSA cryptosystem with $p = 43$ and $q = 59$. 3
 d) Consider an encrypted message "0981 0461". What is the decrypted message if it was encrypted using the RSA cryptosystem with $p = 43$ and $q = 59$. 2

8. a) Alice and Bob develop two new network systems, but they forgot to agree on which security strategy to follow: Strategy 1 (ST#1) or Strategy 2 (ST#2). Their payoffs depend on which strategy they choose and whether they both apply the same one, as shown in the grid below. As always, Alice chooses a row, Bob chooses a column, and Alice's payoff is listed first. 3

		Bob	
		ST#1	ST#2
Alice	ST#1	10, 10	8, 4
	ST#2	4, 8	5, 5

- i) Find the Nash equilibrium of this scenario, if any?
 ii) Does Alice have a strictly dominant strategy? Does Bob? 3
 b) Solve the congruence equation $33X \equiv 38 \pmod{280}$ 3
 c) Consider inserting the keys 21, 8, 3, 13, 44, 28, and 33 into a hash table of length $m=13$ using open addressing with the primary hash function $h(k) = (k+2) \bmod m$. Illustrate the result of inserting these keys using collision avoidance through linear probing. 3

Using section-wise separate answer-scripts, answer any three of the four questions from each of the following two sections. Keep handwriting legible.

SECTION A

Question 1: 2+3+3+1 marks

- (a) Ambiguity is security's enemy – why and how?
(b) Fill in the blanks marked A, B and C in the following with appropriate information/computer security terms, before explaining them in brief:

Consider a simple security policy for a house. No one is allowed in the house unless accompanied by a family member, and only family members are authorized to remove physical objects from the house. An unaccompanied stranger in the house is a security violation. An unlocked back door is a(A)..... A stranger entering through such a door, and removing a television, amounts to(B)..... The attack vector is entry through the unlocked door.

A(C).....here is the existence of an individual motivated to profit by stealing an asset and selling it for cash.

- (c) Write down key roles and responsibilities with the following professions in information/computer security:

- Penetration tester
- Cyber security analyst
- Data security analyst
- Network analyst
- Cyber security auditor
- Cyber security architect

- (d) What is steganography?

Question 2: 3+3+3 marks

- (a) What are the essential ingredients of a symmetric cipher? What is the difference between a block cipher and a stream cipher?

- (b) Explain how cryptanalysis can possibly work for Caesar cipher and one time pad.

- (c) What are the principal elements of a public-key cryptosystem? What are the roles of the public and the private key?

Question 3: 3+2+4 marks

- (a) Outline in general terms an efficient procedure for picking a prime number. The Miller-Rabin test can determine if a number is not prime but cannot determine if a number is prime. How can such an algorithm be used to test for primality?

- (b) Calculating $3^{11} \bmod 17 = X$ is easy, but calculating discrete logarithm $11 = 3^X \bmod 17$ is very difficult – explain.

- (c) Using Fermat's theorem, find $2^{202} \bmod 5$.

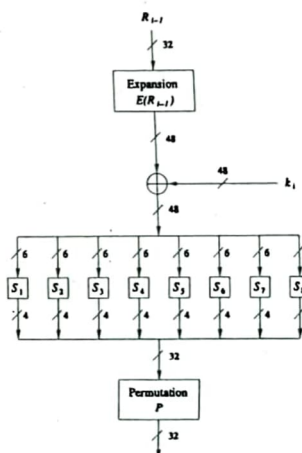
And employ Euler's theorem to find a number X , $0 \leq X \leq 9$, such that X is congruent to 5^{1000} modulo 10.

Question 4: 2+3+4 marks

- (a) Why did IBM include the concept of S-box in Data Encryption Standard (DES)? What is the S_1 -box representation of 37. [The value given at row 3 and col 2 at S_1 table is 08]

- (b) Draw a schematic diagram for 3-DES. Derive an encryption equation for 3-DES, where input text is X and the cipher text is Y .

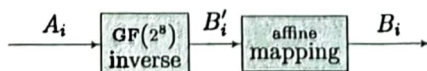
- (c) Give a short description of the components of the f -function in DES as depicted in the following figure.



SECTION B

Question 5: 4.5+4.5 marks

- (a) Describe the two Operations as shown in the following figure



within the Advanced Encryption Standard (AES) S-Box which computes the function $B_i = S(A_i)$. Assume that the S-Box input $A_i = (11000010)_2 = (C2)_{hex}$.

Multiplicative inverse table in $GF(2^8)$ for bytes xy used within the AES box S-Box is as given.

- (b) Sketch and explain the generic model of the digital signature process.

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	00	01	8D	F6	CB	53	7B	D1	E9	4F	29	C0	80	E1	85	C7
1	74	B4	AA	4B	99	2B	60	5F	58	3F	FD	CC	FF	40	EE	B2
2	3A	6E	5A	F1	55	4D	A8	C9	C1	0A	98	15	30	44	A2	C2
3	2C	45	92	6C	F3	39	66	42	F2	35	20	6F	77	BB	59	19
4	1D	FE	37	67	2D	31	F5	69	A7	64	AB	13	54	25	89	09
5	ED	5C	05	CA	4C	24	87	BF	18	3E	22	F0	51	DC	61	17
6	16	5E	AF	D3	49	A6	36	43	F4	47	91	DF	33	93	21	3B
7	79	87	97	85	10	B5	BA	3C	B6	70	00	06	A1	FA	81	82
8	83	7E	7F	80	96	71	88	56	9B	9E	95	D9	F7	02	B9	A4
9	DE	6A	32	6D	D8	BA	84	72	2A	14	9F	8F	F9	DC	89	9A
A	FB	7C	2E	C3	8F	B8	65	48	26	C8	12	4A	CE	E7	D2	62
B	0C	E0	1F	EF	11	75	78	71	A5	8E	76	3D	BD	BC	86	57
C	0B	28	2F	A3	DA	04	0F	A9	27	53	04	1B	FC	AC	8E	66
D	7A	07	AE	63	C5	DB	E2	EA	94	8B	C4	D5	9D	F8	90	6B
E	B1	0D	06	EB	C6	0E	CF	AD	08	4E	D7	83	5D	50	1E	B3
F	5B	23	38	34	68	46	03	8C	DD	9C	7D	AD	CD	1A	41	1C

Question 6: (3+2+2)+2 marks

- (a) Using RSA, encrypt the message $M = 3$, assuming the two primes chosen to generate the keys are $p = 13$ and $q = 7$. You should choose a value $e < 10$. Show your calculations and assumptions.
If Alice used the RSA algorithm in sending the above message $M = 3$ to Bob so that Charlie could not read the message, then what are Alice's and Bob's public keys? Assuming a brute force attack is not possible, explain the steps that Charlie could take to break the encrypted message. (You do not need to perform the calculations, you just need to point out what Charlie needs to calculate and why).
(b) What are the properties a digital signature should have? What requirements should a digital signature scheme satisfy?

Question 7: 2+3+4 marks

- (a) In what order should the signature function and the confidentiality function be applied to a message, and why? What are some threats associated with a direct digital signature scheme?
(b) Write short notes on: SQL injection and social engineering.
(c) SWIFT is a worldwide network used by banks to send each other financial transaction information. Consumers and businesses do not connect directly to SWIFT, only financial institutions can. It has been abused in the last decade, malicious hackers tried to use it to steal US\$1 billion from a central bank.

Ars Technica, a major tech magazine, wrote:

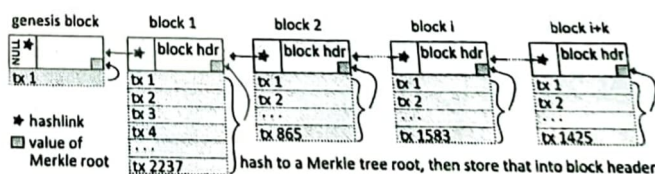
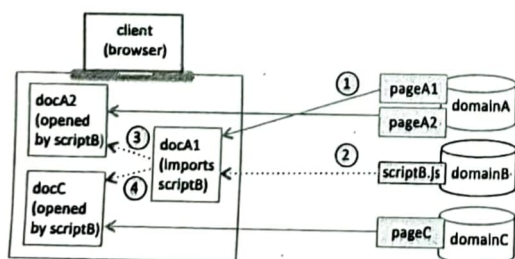
SWIFT's security stems from two major sources. Notionally, it's a private network, and most banks set up their accounts such that only certain transactions between particular parties are permitted. The network privacy means that

it should be hard for someone outside a bank to attack the network, but if a malicious hacker breaks into a bank – as was the case here – then that protection evaporates. The central bank involved has all the necessary SWIFT software and authorized access to the SWIFT network. Any malicious hacker running code within the that central bank also has access to the software and network.

- You've been hired to do a penetration test of some bank's SWIFT gateway. What should your goal be? That is, what concrete, verifiable result would prove that you had successfully hacked your client's SWIFT gateway.
- You of course do not want to damage the SWIFT gateway in any way, and especially not do anything that would leave it open to other attackers. What steps would you take to ensure that there was no damage?

Question 8: 4+4+1 marks

- (a) Explain the *same-origin* policy, preferably using of a full blockchain is given below. following schematic as an illustrative example.



- (b) How is data integrity checked in blockchain? A depiction
(c) How do you distinguish between Ethereum's currency *ether* (ETH) and *bitcoin*?

Department of Computer Science and Engineering
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7TH Semester BSc (Engg.) Examination 2021
CSE 717: INFORMATION SECURITY
Marks: 52.5 Time: 4 hours

[Using section-wise separate answer scripts, answer any three questions from each of the following two sections and all the parts of a question chronologically.]

SECTION A

- A-1/ (a) Derive a risk equation depending on threat agent, probability of attack, and expected loss. 4.75
Draw a related model.
- (b) How do physical and behavioral bio-metrics fail for authentication? How would you evaluate bio-metrics authentication using standard criteria? 3
- (c) Remark on the threats in wireless networks. 1
- A-2/ (a) i. The one-time pad offers complete security – do you agree on this? Defend your answer. 2.75
ii. What are the fundamental difficulties of one-time pad? 2
- (b) Using following Playfair matrix:
- | | | | | |
|---|---|---|-----|---|
| M | F | H | I/J | K |
| U | N | O | P | Q |
| Z | V | W | X | Y |
| E | L | A | R | G |
| D | S | T | B | C |
- Encrypt the message: See you at CSE department 4
- A-3/ (a) i. Calculating $3^{11} \bmod 17 = X$ is easy, but calculating discrete logarithm $11 = 3^X \bmod 17$ is very difficult. Explain above statement. 2.75
ii. Using the extended Euclidean algorithm, find the multiplicative inverse of $1234 \bmod 4321$. 2
- (b) i. Using Fermat's theorem, find $3^{201} \bmod 11$ 2
ii. Use Euler's theorem to find a number x between 0 and 28 with x^{85} congruent to 6 modulo 35. 2
- A-4/ (a) i. For DES, explain the following equation. At the end of the decryption process, we have to reverse the initial permutation:
$$IP^{-1}(R_{16}^d, L_{16}^d) = IP^{-1}(L_0, R_0) = IP^{-1}(IP(x)) = x$$

where x is the plain text that was the input to the DES encryption. 2.75
ii. Draw a block diagram of 3-DES. Also, write down a simple equation to represent subsequent DES encryption for 3-DES. 2
- (b) i. What do you understand by avalanche effect? Write down two families of attacks in DES. 2
ii. Write a program that can encrypt and decrypt using the general Caesar cipher. 2

SECTION B

- B-1/ (a) Draw the Advanced Encryption Standard (AES) encryption block diagram. 4.75
(b) i. Discuss the following properties for cryptographic hash function: *Pre-image resistance*, *Second pre-image resistance*, *Collision resistance*. 2
ii. How can you achieve authentication encryption? 2
- B-2/ (a) Perform encryption and decryption using the RSA algorithm, for $p = 3$, $q = 11$, $e = 7$, $M = 2$. The value of n and cipher-text must be explicitly shown. 2.75
(b) Explain the six steps of Kerberos authentication scheme with appropriate diagram. 2.5
(c) Draw the generic model of digital signature process. 2
(d) State secure hash function requirements. 1.5
- B-3/ (a) i. How is the blockchain related with cryptography? 2.75
ii. What do you understand about bitcoin? 2
(b) Discuss the process of input-output scripts for signature validation for bitcoin. 4
- B-4/ (a) How would you differ between ransomware and social engineering? 2
(b) Discuss the application areas of IPSec. Compare session state and connection state. 2.5
(c) Explain the architecture of IPSec. 2.5
(d) Describe Backdoor, logic bomb, and Trojan horse. 1.75

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[Answer any *three* questions from each of the **Group-A** and **Group-B**. A separate answer script must be used for Group-A and Group-B. Figures in the right-hand margin indicate full marks.]

Group-A

4.75

1. a) Using the following Playfair matrix:

M	F	H	I/J	K
U	N	O	P	Q
Z	V	W	X	Y
E	L	A	R	G
D	S	T	B	C

Encrypt this message: "Must see you over CU playground. Coming now."

4

- b) Using the Vigenère cipher, encrypt the word "explanation" using the key *cse*.

4.75

2. a) Solve the following congruences using the Chinese remainder theorem.

$$X \equiv 1 \pmod{3}$$

$$X \equiv 1 \pmod{4}$$

$$X \equiv 1 \pmod{5}$$

$$X \equiv 1 \pmod{7}$$

- b) (i) Calculating $3^{11} \pmod{17} = X$ is easy, but calculating discrete logarithm $11 = 3^x \pmod{17}$ is very difficult. Explain the above statement. In this aspect, what do you understand by the cyclic group?
- (ii) Find out the distinct remainders for $b^x \pmod{7}$.

2+2

3. a) Using the extended Euclidean algorithm, find the multiplicative inverse of 550 mod 1769.

4.75

- b) (i) Using Fermat's theorem, find $3^{202} \pmod{11}$.

2+2

- (ii) Use Euler's theorem to find a number a between 0 and 9 such that a is congruent to $7^{1000} \pmod{10}$.

4. a) (i) The following questions are related to data encryption standards:

2+2.75

If initial permutation (IP) is there, why do we need IP^{-1} ? What is the S_1 -box representation of 37? [The value given at row 3 and col 2 at S_1 table is 08]

- (ii) Draw the details of the F-function in DES.

- b) (i) For DES, explain the following equation.

At the end of the decryption process

$$IP^{-1}(R_{d_{16}}, L_{d_{16}}) = IP^{-1}(L_0, R_0) = IP^{-1}(IP(x)) = x$$

Where x is the plain text that was the input to the DES encryption.

- (ii) How the concept of the finite field are used in cryptography?

Group-B

5. a) (i) Draw a block diagram of 3-DES. Write down a simple equation to represent DES encryption for 3-DES. 2.75+2
- (ii) What do understand by the avalanche effect? Write down two families of attacks in DES.
- b) Draw the classical Feistel cipher structure for the symmetric block encryption algorithm. 3
- c) What are the differences between a block cipher and a stream cipher? 1
6. a) Perform encryption and decryption using the RSA algorithm, for $p = 3$, $q = 11$, $e = 7$, and $M = 2$. (The value of n and cipher-text must be explicitly shown.) 4.75
- b) (i) In an RSA system, the public key of a given user is $e = 31$, $n = 3599$. What is the private key of this user? 2+2
- (ii) In the RSA public-key encryption scheme, each user has a public key, e , and a private key, d . Suppose Bob leaks his private key. Rather than generating a new modulus, he decides to generate a new public and a new private key. Is this safe?
7. a) Draw the generic model of the digital signature process. 4.75
- b) Find out the 8-bit word related to $x^5 + x^2 + x$. 2
- c) How would you test a number $n = 29$ is a prime or not using the Miller-Rabin algorithm? Show the steps clearly. 2
8. a) Determine the benefits of IPSec. What are the differences between transport mode and tunnel mode? 2.25
- b) What are the general services defined by RFC4301 for IPSec? 2.25
- c) Discuss the application areas of IPSec. Compare session state and connection state. 2.25
- d) Explain the architecture of IPSec. 2